

Application Layer Protocols in Computer Networks: Technical Briefing

Summary

The **Application Layer** is the interface between users and network services. It supports thousands of services, but key protocols widely referenced in networking and exams like GATE include **SMTP, FTP, HTTP, POP3/IMAP, and DNS**.

Key observations:

- **Transport Layer Dependency:** Most protocols use **TCP** for reliability, while a few (DNS, TFTP, DHCP) prefer **UDP** for speed.
- **State Management:** Protocols can be **stateless** (HTTP, SMTP, DNS) or **stateful** (POP3, IMAP, FTP).
- **Connection Modes:** Include **persistent** (IMAP, HTTP/1.1, SMTP) and **non-persistent** (HTTP/1.0, FTP data).
- **Push vs Pull:** SMTP “pushes” emails; POP3/IMAP “pull” emails.
- **DNS Hierarchy:** Root → TLD → Authoritative → Local, with iterative or recursive queries.

1. Electronic Mail Protocols

SMTP – Simple Mail Transfer Protocol

- **Purpose:** Push emails from client to server and server to server.
- **Port:** TCP 25
- **Characteristics:** Text-based, connection-oriented, persistent, in-band, stateless.
- **Operational Stages:** Used twice per email—client-to-server and server-to-server.

MIME – Multipurpose Internet Mail Extensions

- Enables SMTP to handle non-ASCII content (images, audio, video) by converting binary to text.

POP3 vs. IMAP4

Feature	POP3 (Post Office Protocol v3)	IMAP4 (Internet Mail Access Protocol v4)
Port	110 TCP	143 TCP
Device Focus	Single device	Multiple device synchronization
Partial Download	Not allowed	Allowed (e.g., headers only)
Folder Management	Limited	Full folder management
State	Stateful (delete/keep modes)	Stateful
Complexity	Simple	More complex

Summary: IMAP is superior for multi-device access, while POP3 is simpler for single-device downloads.

2. Domain Name System (DNS)

Function: Resolves human-readable names to IP addresses.

Hierarchy:

1. **Root Servers:** 13 worldwide; critical for internet functionality.
2. **TLD Servers:** Handle domain suffixes (.com, .org, .edu).
3. **Authoritative Servers:** Hold actual mappings of domain names to IPs.
4. **Local DNS Server:** Caches frequently accessed domains for speed.

Query Types:

- **Iterative:** Server refers the client to another server.
- **Recursive:** Server resolves query completely and responds to client.

Transport and Port:

- **Port:** 53
- **Protocol:** UDP by default (fast), TCP if query >512 bytes.

3. File Transfer Protocol (FTP)

- **Purpose:** Transfer files reliably between computers.
- **Dual Connection Mechanism (Out-of-Band):**
 - **Control Connection:** TCP 21, persistent, for commands.
 - **Data Connection:** TCP 20, non-persistent, for file transfer.

Transmission Modes:

1. **Stream Mode:** Continuous byte stream.
2. **Block Mode:** Data sent in blocks with headers.
3. **Compressed Mode:** Compresses large files.

4. Hypertext Transfer Protocol (HTTP)

- **Purpose:** Access web resources.
- **Port:** TCP 80
- **Characteristics:** In-band, stateless.

Connection Types:

- **Non-Persistent (HTTP/1.0):** Each object requires a separate connection.
- **Persistent (HTTP/1.1):** Single connection can transfer multiple objects, reducing overhead.

5. Comparative Protocol Analysis

Port and Transport Summary:

Protocol	Port(s)	Transport
DNS	53	UDP/TCP
HTTP	80	TCP
FTP	20 (Data), 21 (Ctrl)	TCP
SMTP	25	TCP
POP3	110	TCP
IMAP	143	TCP

Telnet	23	TCP
DHCP	67/68	UDP
TFTP	69	UDP

Operational Modes:

Protocol	State	Band Type	Connection Mode
DNS	Stateless	In-band	Connectionless
HTTP	Stateless	In-band	Non-persistent (1.0) / Persistent (1.1)
SMTP	Stateless	In-band	Persistent (Push)
POP3	Stateful	In-band	Persistent (Pull)
IMAP	Stateful	In-band	Persistent (Pull)
FTP	Stateful	Out-of-band	Control (Persistent) / Data (Non-persistent)

6. Key Commands (Examination Focus)

- **HTTP:** GET, POST, HEAD, PUT, DELETE
- **SMTP:** HELO, MAIL FROM, RCPT TO, DATA, QUIT, VERIFY
- **FTP:** USER, PASS, PORT, RETR, STOR, QUIT, CWD
- **POP3:** USER, PASS, STAT, LIST, RETR, DELE